



9/1/2026

N + - Network Fundamentals & Building Networks.

Hands-on Subnetting.

TASK 1.

- ❖ Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

➤ **Dividing a /24 Network into 4 Equal Subnets:**

Given: 192.168.10.0/24

To create **4 equal subnets**, 2 bits are borrowed from the host portion:

$2^2 = 4$ subnets

New subnet mask: **/26 = 255.255.255.192**

Subnet	Subnet Mask	Network Address	Broadcast Address
1	255.255.255.192	192.168.10.0	192.168.10.63
2	255.255.255.192	192.168.10.64	192.168.10.127
3	255.255.255.192	192.168.10.128	192.168.10.191
4	255.255.255.192	192.168.10.192	192.168.10.255

TASK 2.

- ❖ For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device, Make sure not to use the network or broadcast address for any device.

➤ **Assigning Valid IP Addresses:**

For this task, I selected the first subnet: **192.168.10.0/26**.

↗ **Network Address:** 192.168.10.0

↗ **Broadcast Address:** 192.168.10.63

↗ **Valid Host Range:** 192.168.10.1 – 192.168.10.62

Device	Assigned IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile	192.168.10.5

All five IP addresses are valid host addresses and do not use the network or broadcast address.

TASK 3.

❖ Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

➤ Subnet Plan Diagram:

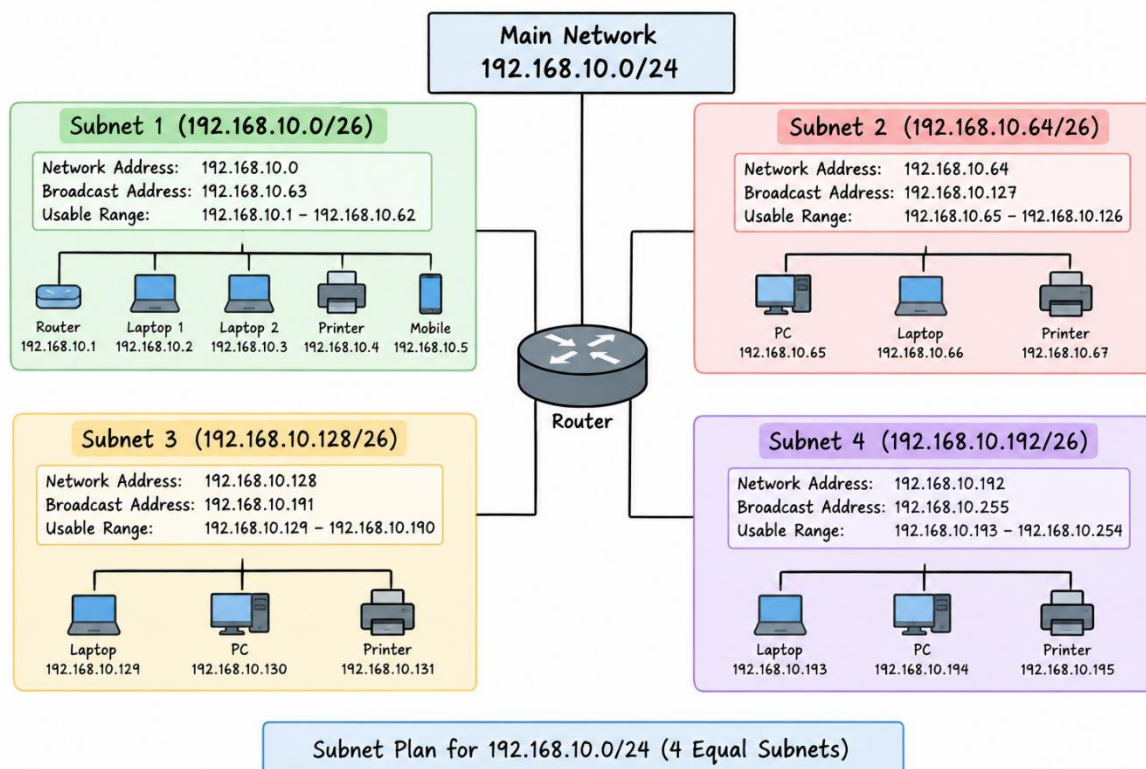


Figure : Subnet Plan for 192.168.10.0/24.

⌘ **Subnet 1:** 192.168.10.0/26 — Usable IPs: 192.168.10.1–62 — Devices: Router, Laptop 1, Laptop 2, Printer, Mobile

⌘ **Subnet 2:** 192.168.10.64/26 — Usable IPs: 192.168.10.65–126

⌘ **Subnet 3:** 192.168.10.128/26 — Usable IPs: 192.168.10.129–190

⌘ **Subnet 4:** 192.168.10.192/26 — Usable IPs: 192.168.10.193–254

TASK 4.

- ❖ Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

➤ Subnetting Calculation and IP Assignment:

⌘ Step 1: Calculate the Subnet Mask

The original network is **192.168.10.0/24**.

We need **4 equal subnets**, so we borrow **2 bits** because:

$$2^2 = 4 \text{ subnets}$$

Therefore, the new prefix is:

$$/24 + 2 = /26$$

Subnet Mask: **255.255.255.192**

⌘ Step 2: Calculate Hosts per Subnet

With **/26**, there are **6 bits left for hosts**.

$$2^6 = 64 \text{ total IP addresses}$$

Two addresses are reserved for the network and broadcast addresses:

$$64 - 2 = 62 \text{ usable hosts per subnet}$$

⌘ Step 3: Assign IP Addresses

The subnet block size is **64**, so the four subnets are:

Subnet	Network Address	Usable IP Range
Subnet 1	192.168.10.0/26	192.168.10.1 – 192.168.10.62

Subnet 2	192.168.10.64/26	192.168.10.65 – 192.168.10.126
Subnet 3	192.168.10.128/26	192.168.10.129 – 192.168.10.190
Subnet 4	192.168.10.192/26	192.168.10.193 – 192.168.10.254

For **Subnet 1**, the five devices are assigned:

- ➦ Router → **192.168.10.1**
- ➦ Laptop 1 → **192.168.10.2**
- ➦ Laptop 2 → **192.168.10.3**
- ➦ Printer → **192.168.10.4**
- ➦ Mobile → **192.168.10.5**